Comparing social needs for transport, gaps in transit service and the Mobility Index for the west of Ireland

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Introduction

Research Background

Many indexes, metrics and other indicators have been developed by researchers, practitioners and others to examine transport, land use and related topics. Examples include: *Walk Score* and *Transit Score*,[1] which rate a location out of 100 for walkability and transit accessibility respectively; metrics within the *Transit Capacity and Quality of Service Manual (TCQSM)*;[2] and the UK's Public Transport Accessibility Level (PTAL) approach and Public Transport Accessibility Index (PTAI).[3]

What may sometimes be lacking is comparison of the results from one set of indicators to another other. In part this can be because of the variability in data availability. Censuses often provide a key resource for understanding transport-related issues, yet collection practices used, questions asked and other elements tend vary significantly from one jurisdiction to another. Likewise, transport agencies, government departments and other sources of data might have different approaches with respect to data sharing, open data policies or variability in what datasets they collect or produce in the first place. Hence, not all indicators can be calculated in all jurisdictions.
That said,
many indicators have been developed for specific contexts. An example is the recently reported a Mobility Index (MI)[4] for rural centres in the west of Ireland, developed because other MIs tend to be for more urbanised locations where origins and destinations tend to be closer together.

Finally, some indicators are difficult or impossible to calculate independently. Some are closed-source and can only be obtained from a single entity[5]. Others may just be challenging to calculate due to a lack of, or free access to, software tools.

While censuses and the collection of many other datasets at local, regional and national levels appear likely to remain specific to their own geographic contexts, there have been some recent developments in technology and transport practices that allow for universal (or at least internationally-applicable) tools and approaches. Wong (2013)[6] developed software to calculate many of the *TCQSM* indicators from

[1] Walk Score, "Transit Score Methodology," 2023, `https://www.walkscore.com/transit-score-methodology.shtml`.

[2] Kittleson & Associates et al., *Transit Capacity and Quality of Service Manual, Third Edition*, Third Edition, TCRP Report 165 (Washington DC: Transportation Research Board; Transportation Research Board, 2013), `http://www.trb.org/Main/Blurbs/169437.aspx`.

[3] E Saghapour, S Moridpour, and R Thompson, "Public Transport Accessibility in Metropolitan Areas: A New Approach Incorporating Population Density," *Journal of Transport Geography* 54 (2016): 273–85.

[4] Helen McHenry, Amaya Vega, and Catherine Swift, "Understanding Mobility in Rural Towns: Development of a Mobility Index for the West of Ireland," *Transportation Research Procedia* 72 (2023): 3730–37, doi:`https://doi.org/10.1016/j.trpro.2023.11.545`.

[5] e.g. WalkScore and Transit Score cannot be independently calculated and the exact algorithm remains unpublished

[6] James Wong, "Leveraging the General Transit Feed Specification for Efficient Transit Analysis," *Transportation Research Record* 1, no. 2338 (2013): 11–19, doi:10.3141/2338-02.

General Transit Feed Specification (GTFS) datasets, which is freely available online. This is possible because of the development and increased availability of GTFS feeds, which were originially introduced to support the provision of transit directions on the Google Maps platform. This allows transit operators to publish their static timetable data in an open, simple and text-based format, which has been adopted by over 10,000 agencies[7] Hence, software tools that take GTFS data as inputs can generally be adapted easily and rapidly to many systems and locations, as in how Wong (2013) compared TCQSM metrics across 50 USA transit operators.

[7] MobilityData, *General Transit Feed Specification (GTFS)*, undated, `%7Bhttps://gtfs.org/%7D`.

Another such recently developed software tool is the *gtfssupplyindex* R package,[8] which builds on other GTFS-related R packages including *gtfstools*;[9] and *tidytransit*.[10] The *gtfssupplyindex* can be used to calculate a transit Supply Index (SI), developed in research related to identifying gaps in public transport services on the basis of social needs for transport reported in Currie (2010).[11] The SI equation is:

[8] James Reynolds, Graham Currie, and Yanda Qu, "Social Needs for Transport and Gaps in Transit Service: New GTFS Tools," *Submitted to Australasian Transport Research Forum. November, Auckland, NZ.*, under review.

[9] Daniel Herszenhut et al., *Gtfstools: General Transit Feed Specification (GTFS) Editing and Analysing Tools*, 2025, `https://ipeagit.github.io/gtfstools/`.

[10] Flavio Poletti et al., *Tidytransit: Read, Validate, Analyze, and Map GTFS Feeds*, 2024, `https://github.com/r-transit/tidytransit`.

[11] Graham Currie, "Quantifying Spatial Gaps in Public Transport Supply Based on Social Needs," *Journal of Transport Geography* 18, no. 1 (2010): 31–41.

$$SI_{area,time} = \sum_{i=1}^{n} \frac{Area_{Bi}}{Area_{area}} SL_{i,time}$$

where:

- $SI_{area,time}$ is the Supply Index for the *area of interest* and a given period of time;
- $Area_{Bi}$ is the amount of the buffer zone for each stop (i) that is within the *area of interest*[12];
- $Area_{area}$ is the area of each *area of interest*; and
- $SL_{i,time}$ is the number of transit arrivals for each stop (i) within the given time period.

[12] In Currie, ibid this buffer zone was based on a radius of 400 metres for bus and tram stops, and 800 metres for railway stations. The same definition is used here.

Currie (2010) also used a Composite Social Needs Index, based mostly on census data including measures related to income, age, car ownership and other indicators related to the extent to which those living in each area of interest might have higher or lower needs for public transport.
The final step of the Currie (2010) social needs-gap approach involved identifying those areas of interest that had both high social needs, but low or zero transit supply levels.

This approach, and its potential application to policy-making, route design and service planning was suggested to be "substantially more useful than the presentation of anecdotal evidence, which is the most common means of identifying transport needs in local transport studies throughout the world".[13]

[13] Ibid.

A feature of the SI equation is that scores can be summed across areas of interest so as to provide an aggregate score for a larger district, region or city. The SI also had the advantage of including the frequency and coverage of transit services, meaning that the score reflects the accessibility 'to' public transport for those living within an area, but also how accessible the area is 'by' transit.

The development of the Currie (2010) needs-gap approach predated the GTFS, and hence the analysis required bespoke cleaning and processing of data obtained directly from the transit agency. Reynolds, Currie, and Qu,[14] however, reported the development of the *gtfssupplyindex*, which allows SI scores to be calculated directly from GTFS files, and results from applying this to Melbourne, Australia. Beyond this, however, the *gtfssupplyindex* package has not yet been applied to other locations, or indeed rural geographies. Also unclear is how such results might compare to other transport-related indicators, including the Mobility Index for rural areas developed by McHenry et al (2023)[15]

In the following sections, therefore, this paper reports the application of the *gtfssupplyindex* package to the west of Ireland, adopting the same case and context as McHenry et al (2023). Results from a social needs-gap analysis are compared to those of McHenry et al (2023) so as to provide insights into the differences between these indicators. This research is anticipated to be part of a larger programme that would continue to apply different indicators and approaches to this case, so as to allow deeper understandings of both transport in the west of Ireland, but also of how well existing indicators and other analysis techniques might allow researchers, practitioners and policy-makers to understand and improve rural transport networks.

Methodology

The *areas of interest* adopted in this study are the 2022 definition of *Small Areas*, as used in the Irish Census[16] GTFS data for this paper was sourced from Transport for Ireland[17] on October 1, 2025[18]. The time period selected was the seven days starting October 1, 2025.

The same area of interest categorisation approach as in Currie (2010) and Reynolds et al (2025) was adopted[19]

[14] "Social Needs for Transport and Gaps in Transit Service."

[15] McHenry, Vega, and Swift, "Understanding Mobility in Rural Towns."

[16] However, data from the Irish Census is reported seperately for Galway City and the rest of County Galway("Galway City" is included as an additional county in the COUNTY_ENGLISH field), whereas in this analysis Small Areas within Galway City are reported as part of County Galway.

[17] "Public Transport Data" (National Transport Authority, undated), `https://www.transportforireland.ie/transitData/PT_Data.html`.

[18] Selecting the "All GTFS Operators" feed

[19] Transit supply is grouped into seven categories, based on first allocating all areas of interest with a SI score of zero into "Zero Supply". The average SI score of the remainder is then used to split the rest of the areas of interest, which are then further subdivided into three even groups ("Very Low", "Low", "Below average") below and three even groups above ("Above average", "High", "Very High) the mean. A

Results

SI scores and transit supply categories

Table 1 shows summary statistics for SI scores, and by transit supply category.

Table 1: Summary statistics; SI score by transit supply category, by Small Area.

Characteristic	Overall N = 3,666	Transit supply category						
		Zero Supply N = 1,172	Very Low N = 683	Low N = 683	Below average N = 683	Above average N = 149	High N = 148	Very High N = 148
SI								
Min	0.0	0.0	0.0	2.3	37.3	418.5	986.6	2,178.9
Q1	0.0	0.0	0.1	4.6	64.2	513.1	1,286.1	2,610.6
Mean	281.1	0.0	0.7	13.6	144.3	665.4	1,586.6	3,973.2
Q3	73.0	0.0	1.2	21.3	197.1	812.9	1,951.6	4,140.4
Max	11,860.5	0.0	2.3	37.2	411.3	978.1	2,177.0	11,860.5
Sum	1,030,354.4	0.0	507.8	9,282.6	98,566.1	99,148.7	234,815.0	588,034.3

The lower and upper SI score thresholds for each of the transit supply categories are shown in Table 2, along with the number of Small Areas in each category.

Table 2: Transit supply category groupings, SI score thresholds and number of Small Areas in each category.

Category	Lower	Upper	Areas
Zero Supply	0	0	1,172
Very Low	0	2	683
Low	2	37	683
Below average	37	412	683
Above average	412	978	149
High	978	2,180	148
Very High	2,180	11,900	148

Some 1,172 of the 3,666 Small Areas in the west of Ireland (32%) have SI scores of zero, implying that there is no transit service at all. Of those Small Areas that had at least some transit, 2,049 (56% of the total) had SI scores below the average score (412), with the remaining 445 (12% of the total Small Areas in the west of Ireland.

A map and the distribution of transit Supply Index (SI) categories across the west of Ireland is shown in Figure 1. This Figure also shows the population living within Small Areas in each category, including by County.

In general, Figure 1 indicates that large parts of the west of Ireland have Very Low levels of transit or no transit at all. As might be expected, those Small Areas where SI scores are higher tend to be in population centres. However, of the total population across the west of Ireland (886,385) 87% (770,476) live in Small Areas where the SI score is below the average (of those Small Areas with a non-zero SI

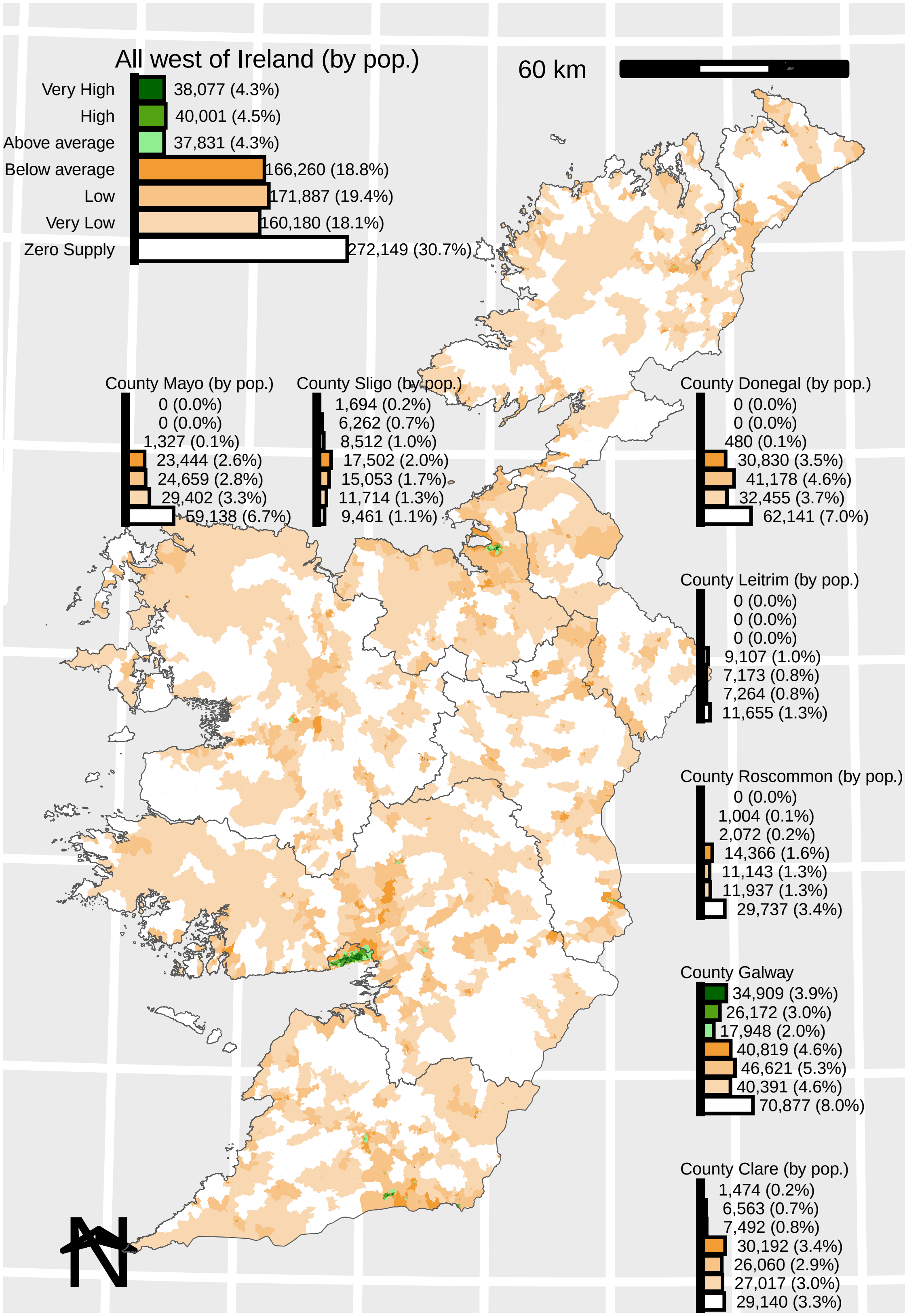

Figure 1: Transit Supply Index categories across the west of Ireland, by Small Area and by population

score). This includes the 272,149 people (31%) who live in Small Areas with no transit at all. The smaller bar charts in Figure 1 indicate that the largest shares of the total population of the west of Ireland are the 70,877, 62,141 and 59,138 people living in County Galway, County Donegal and County Mayo in Small Areas with no transit at all.

There is a statistically significant variation across Counties in the share of Small Areas in each transit supply category ($\chi^2(36) = 700.76$, $p < .001$). Table 3 summarises the Small Area populations in each County by transit supply category.

Table 3: Share of populations by transit supply catergory and County.

County	Transit Supply category							Total
	Zero Supply	Very Low	Low	Below av.	Above av.	High	Very High	
GALWAY	25.5%	14.5%	16.8%	14.7%	6.5%	9.4%	12.6%	100.0%
	(70,877)	(40,391)	(46,621)	(40,819)	(17,948)	(26,172)	(34,909)	(277,737)
DONEGAL	37.2%	19.4%	24.6%	18.5%	0.3% (480)	0.0% (0)	0.0% (0)	100.0%
	(62,141)	(32,455)	(41,178)	(30,830)				(167,084)
MAYO	42.9%	21.3%	17.9%	17.0%	1.0%	0.0% (0)	0.0% (0)	100.0%
	(59,138)	(29,402)	(24,659)	(23,444)	(1,327)			(137,970)
CLARE	22.8%	21.1%	20.4%	23.6%	5.9%	5.1%	1.2%	100.0%
	(29,140)	(27,017)	(26,060)	(30,192)	(7,492)	(6,563)	(1,474)	(127,938)
ROSCOMMON	42.3%	17.0%	15.9%	20.4%	2.9%	1.4%	0.0% (0)	100.0%
	(29,737)	(11,937)	(11,143)	(14,366)	(2,072)	(1,004)		(70,259)
SLIGO	13.5%	16.7%	21.4%	24.9%	12.1%	8.9%	2.4%	100.0%
	(9,461)	(11,714)	(15,053)	(17,502)	(8,512)	(6,262)	(1,694)	(70,198)
LEITRIM	33.1%	20.6%	20.4%	25.9%	0.0% (0)	0.0% (0)	0.0% (0)	100.0%
	(11,655)	(7,264)	(7,173)	(9,107)				(35,199)
Total	30.7%	18.1%	19.4%	18.8%	4.3%	4.5%	4.3%	100.0%
	(272,149)	(160,180)	(171,887)	(166,260)	(37,831)	(40,001)	(38,077)	(886,385)

Table 3 shows that that 43% of people living in Mayo, 42% in Roscommon and 37% in Donegal live in Small Areas with Zero Supply, compared to just 13% in Sligo, 23% in Clare and 26% in Galway. At the other end of the spectrum, some 28% of those in Galway live in Small Areas categorised as having Above average, High or Very High transit service levels, 23% of those in Sligo and12% of those in Clare. In contrast, there are no Small Areas categorised as having Above average, High or Very High transit service levels in Leitrim, while only 0.3% of the Donegal population and 1.0% of those in Mayo live in Small Areas with Above average, High or Very High transit service levels. This is, however, perhaps not surprising given that there are major population centres in the west of Ireland in County Galway (Galway City), County Clare (Ennis, Shannon and parts of Limerick City) and County Sligo (Sligo). However, as discussed further in the next section, Letterkenny (County Donegal), Castlebar and Ballina (County Mayo), and some other urban areas appear to have lower levels of transit supply than is the case for other places with similar (or smaller) populations.

Urban areas Approximately 50% of the west of Ireland's population (close to 435,000 people) live in Small Areas that are within one of the 242 named 'urban areas' within the west. Figure 2 shows the total population and aggregate SI scores for each urban area.

There is a statistically significant relationship between population and aggregate SI score for urban areas in the west of Ireland ($r(243) = .90$, $p < .001$). However, as shown in Figure 2, there are outliers at both ends of the spectrum of both population and aggregate SI.

Needs Index

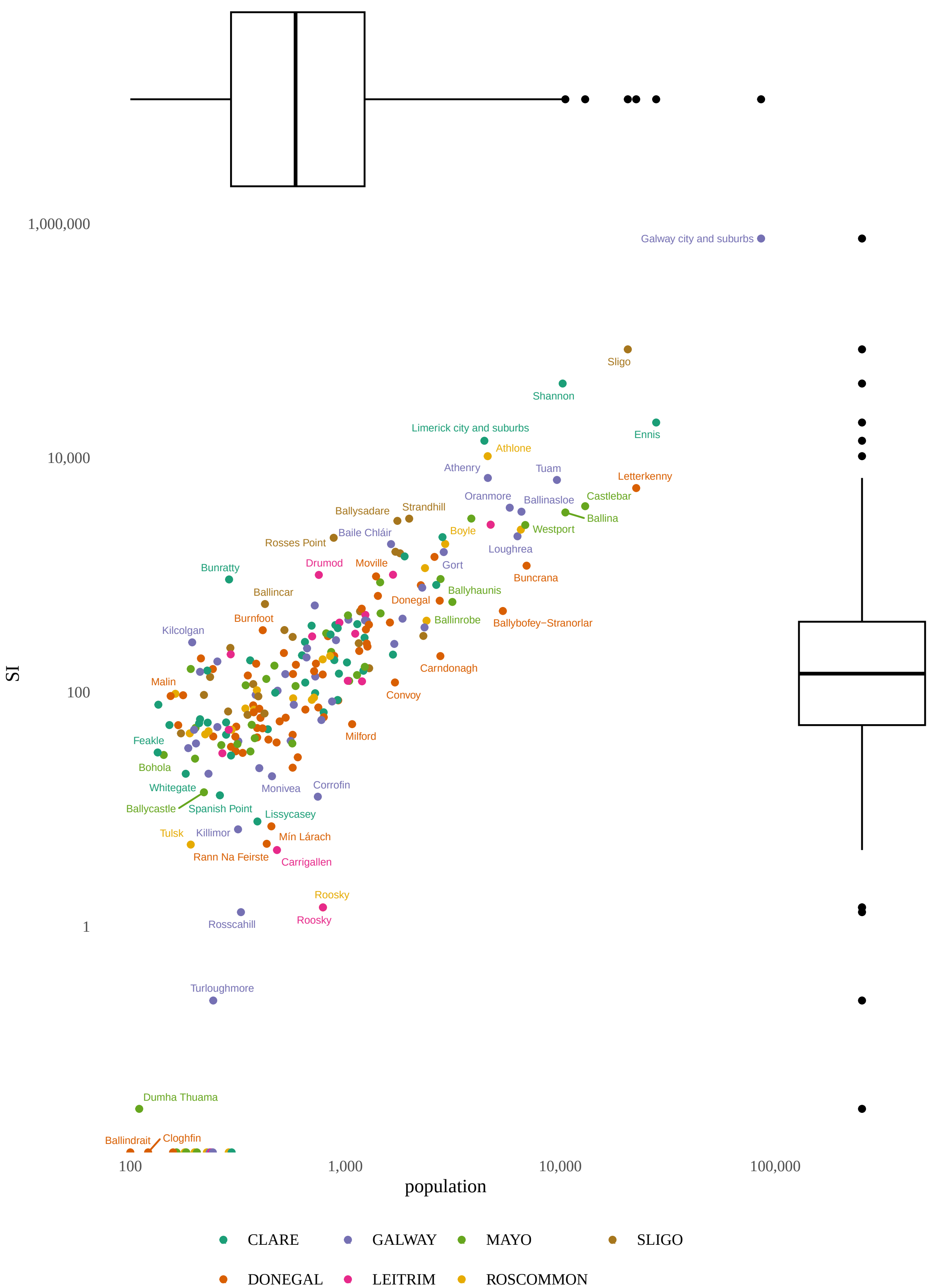

Figure 2: West of Ireland pop. vs SI, aggregated by urban areas